Project:

Customer Segmentation using K-Means Clustering

Problem Statement

Businesses often struggle to understand different types of customers. Without proper segmentation, marketing strategies become inefficient.

This project aims to group customers into meaningful segments based on their income and spending behavior using K-Means clustering.

Objective

- To identify distinct customer groups
- To analyze customer purchasing behavior
- To help businesses target the right audience
- To improve decision-making using data

STEP 1: Import Libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

STEP 2: Load Dataset

In [2]:
```
df = pd.read_csv("Realistic_Mall_Customers.csv")
df.head()
```

Out[2]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Female	38	48.88	53
1	2	Female	41	122.13	25
2	3	Male	18	54.29	40
3	4	Female	38	121.40	19
4	5	Male	27	52.97	28

STEP 3: Data Understanding

In [3]:
```
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              2000 non-null   int64
 1   Gender                  2000 non-null   str
 2   Age                     2000 non-null   int64
 3   Annual Income (k$)      2000 non-null   float64
 4   Spending Score (1-100)  2000 non-null   int64
dtypes: float64(1), int64(3), str(1)
memory usage: 78.3 KB
```

Out[3]:

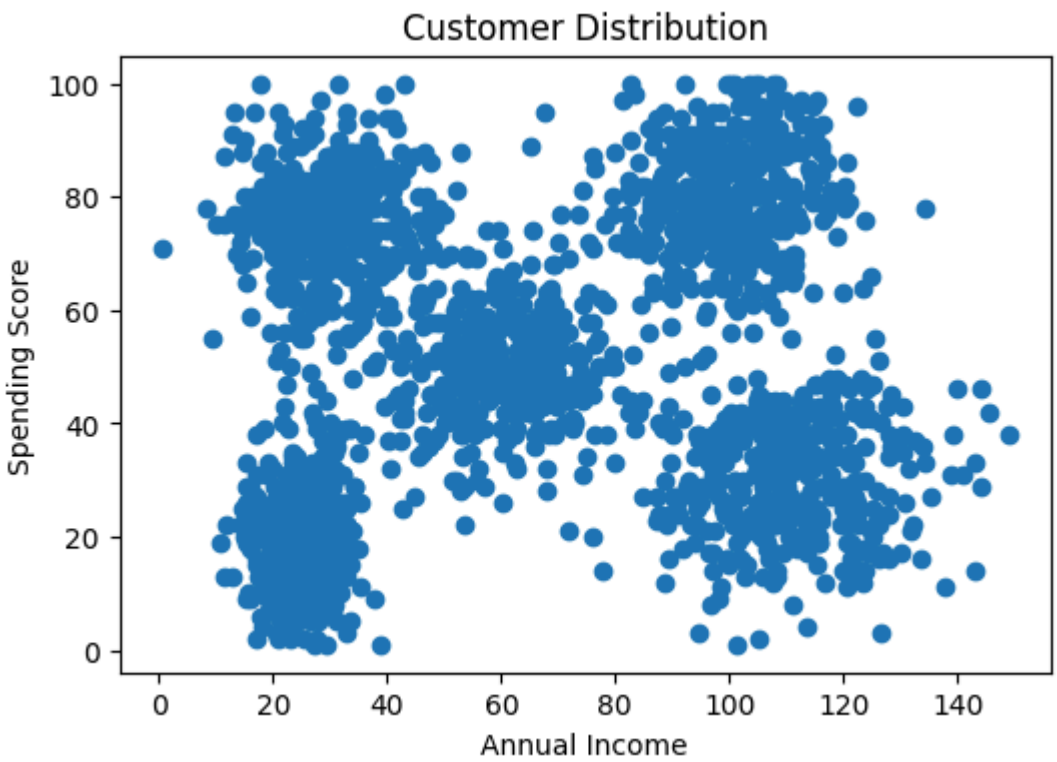
```
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64
```

- Dataset is clean with no missing values and appropriate data types.

STEP 4 : EDA

In [4]:

```
plt.figure(figsize=(6,4))
plt.scatter(df['Annual Income (k$)'], df['Spending Score (1-100)'])
plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Distribution")
plt.show()
```



STEP 5 : Feature Selection

- Selecting relevant features for clustering.

In [5]:
```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

STEP 6 : Feature Scaling

- Scaling ensures all features contribute equally to distance calculation.

In [6]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

STEP 7 : Elbow Method

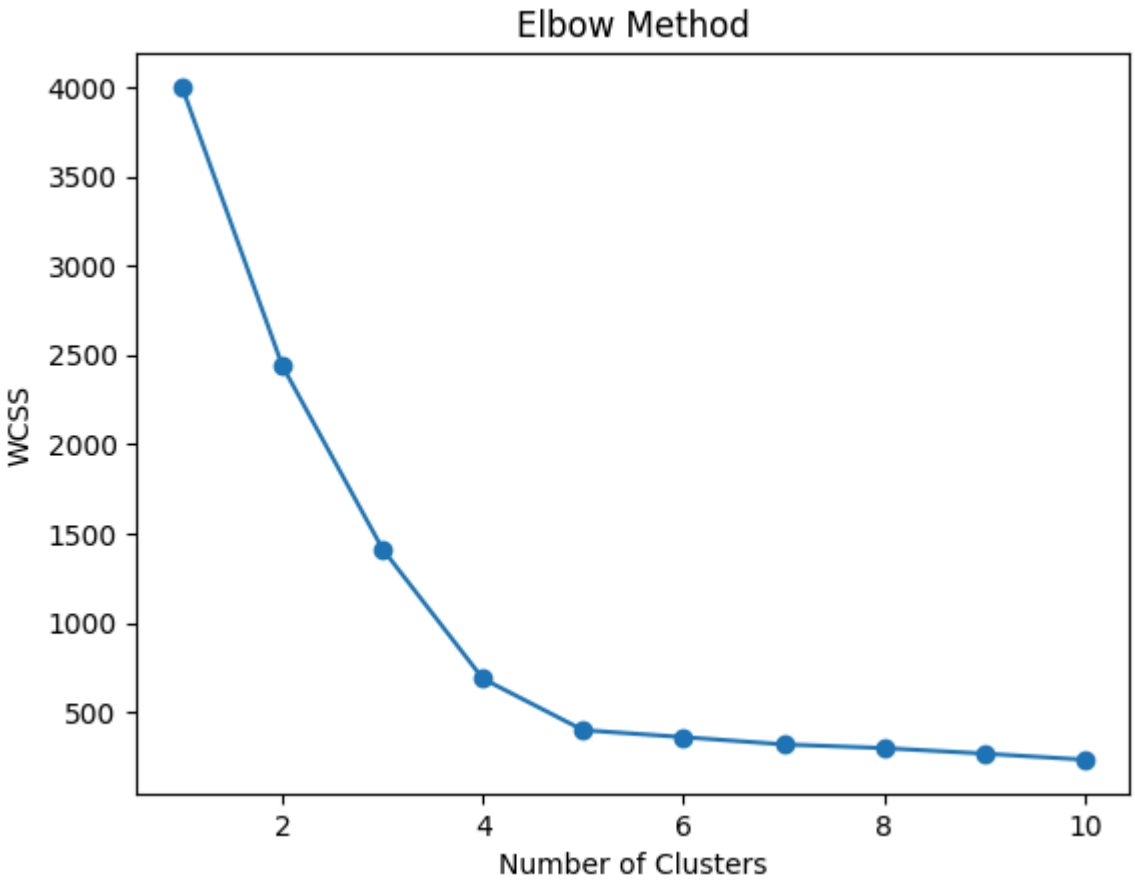
- Used to determine the optimal number of clusters (K).

In [7]:
```
from sklearn.cluster import KMeans

wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
```

In [8]:
```
plt.plot(range(1,11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```



- The optimal K is selected at the "elbow point" where WCSS decreases slowly.

STEP 8 : Apply K-Means

In [9]:
```
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
```

STEP 9 : Visualization of Clusters

- Visualizing clusters to interpret customer segments.

In [10]:
```
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=y_kmeans, cmap='viridis')

plt.scatter(kmeans.cluster_centers_[:,0],
            kmeans.cluster_centers_[:,1],
            s=200, c='red', label='Centroids')

plt.title("Customer Segments")
plt.xlabel("Income (Scaled)")
plt.ylabel("Spending Score (Scaled)")
plt.legend()
plt.show()
```

Insights from Customer Segmentation

- The K-Means algorithm successfully divided customers into 5 distinct clusters, clearly showing different spending behaviors based on income levels.

Cluster 1: Low Income – High Spending

(Left top cluster)

Characteristics:

Low income High spending score

Insight:

These customers tend to spend a large portion of their income. They may be impulsive buyers or attracted by offers and discounts.

Business Strategy:

Provide discounts, coupons, and deals Target with promotional campaigns

Cluster 2: High Income – High Spending

(Right top cluster)

Characteristics:

High income High spending

Insight:

These are premium customers and the most valuable segment for the business.

Business Strategy:

Offer premium services Loyalty programs Exclusive memberships

Cluster 3: Medium Income – Medium Spending

(Center cluster)

Characteristics:

Average income Average spending

Insight:

These are regular customers who provide stable revenue.

Business Strategy:

Maintain engagement Offer personalized recommendations

Cluster 4: Low Income – Low Spending

(Bottom left cluster)

Characteristics:

Low income Low spending

Insight:

These customers contribute less to revenue and are low-priority customers.

Business Strategy:

Minimal marketing investment Focus only on basic engagement

Cluster 5: High Income – Low Spending

(Bottom right cluster)

Characteristics:

High income Low spending

Insight:

These are potential customers who have the capacity to spend more but currently do not.

Business Strategy:

Target with personalized marketing Offer premium product recommendations Convert them into high spenders

Conclusion

The K-Means clustering algorithm effectively segmented customers into meaningful groups based on their income and spending behavior.

The model identified premium, regular, low-value, and potential customers, allowing businesses to design targeted marketing strategies.

This segmentation can help improve customer engagement, increase sales, and optimize resource allocation.